

World-Model Inspired Emotion-aware Token Refinement for Training-Free Multimodal Emotion Recognition

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Abstract

Multimodal Large Language Models (MLLMs) show promise for Multimodal Emotion Recognition (MER) but often remain unreliable because sparse emotional cues could be easily overwhelmed and affected by redundant context. While fine-tuning is effective, it is usually costly when using large models. Training-free methods like chain-of-thought reasoning provide a practical alternative, but they mostly rely on heuristic prompting to influence the model behaviors and do not explicitly focus on emotion relevant tokens internally, which would allow decision-relevant emotional tokens to be diluted by environmental noise, resulting in unstable predictions. To address this limitation without training, we rethink MER from a world-model perspective that treats emotion as a latent state inferred from noisy and redundant multimodal observations. Under frozen parameters, this view suggests that robustness depends on constraining why and how tokens contribute to inference. Based on this insight, we propose **WETR** (World-Model inspired Emotion-aware Token Refinement), a training-free, plug-and-play regulator that reshapes token usage through two mechanisms: Noise-suppressed Token Selection (NTS), which suppresses redundant intra-modal noise, and State-strengthened Token Reweighting (STR), which amplifies decision-relevant emotional tokens. Experiments on multiple MER benchmarks demonstrate that WETR consistently improves accuracy and stability under frozen parameters, which also improves token-level interpretability.

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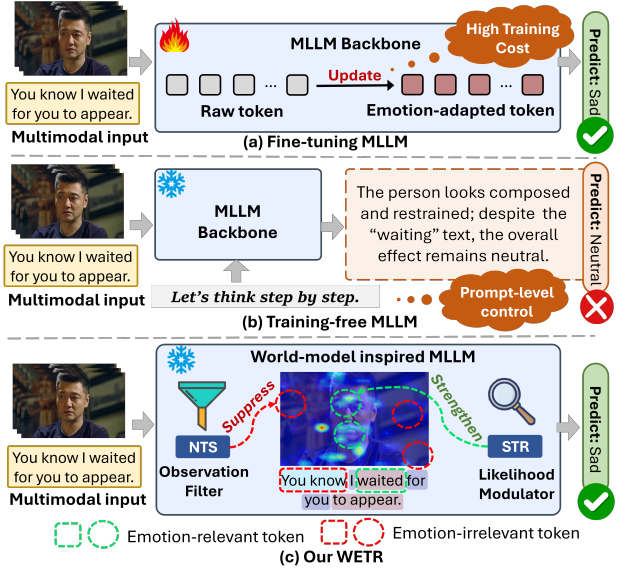


Figure 1. Different methods to improve MER with MLLMs. (a) Fine-tuning updates parameters to obtain emotion-adapted tokens. (b) Training-free prompting may be distracted by emotion-irrelevant context (predict neutral). (c) WETR refines tokens at inference by suppressing irrelevant tokens (NTS) and strengthening emotion cues (STR), correcting to sad.

1. Introduction

Multimodal large language models (MLLMs) have attracted increasing attention for their strong cross-modal understanding and reasoning abilities (Li et al., 2024; Chen et al., 2024b; Liu et al., 2024; Huang et al., 2024), achieving impressive performance on tasks such as visual question answering and visual reasoning (Zhang et al., 2025). Despite this progress, applying MLLMs to multimodal emotion recognition (MER) remains challenging: emotional expressions are often subtle and sparsely distributed (e.g., small facial muscle changes in videos, contrastive turns in text, etc.) (Cheng et al., 2024; Yang et al., 2024), while real-world data typically contains substantial redundant and irrelevant context or background noise. As a result, MLLMs may struggle to attend to decision-relevant emotional cues. Consequently, even if MLLMs could generate a seemingly reasonable emotional response, its decision may not be grounded in the truly informative emotional evidence, leading to unstable and unreliable emotion reasoning.

Recent studies have begun to address the difficulty MLLMs face in leveraging decision-relevant emotional cues for MER (Zhang et al., 2024a; Zhao et al., 2024; Zhang et al., 2024b). Existing approaches largely fall into two categories: fine-tuning-based methods and training-free inference enhancements. Fine-tuning approaches typically train specialized encoders to better capture emotional signals across modalities (Cheng et al., 2024), or employ auxiliary MLLMs to generate emotion-aware textual descriptions (Xie et al., 2024), followed by supervised training on emotion-related datasets. While effective, such methods incur substantial training costs, limit scalability, and are often impractical in realistic deployments that depends on frozen large-scale MLLMs. Motivated by these constraints, an increasing number of studies explore training-free inference-time strategies, including prompt engineering, chain-of-thought reasoning, and self-reflection (Zhang et al., 2024a; Fang et al., 2025). Although these methods can guide MLLMs toward emotion-related information, they primarily operate at the input or output level and continue to rely on the pretrained token-level attention and aggregation mechanisms. Without explicitly regulating how decision-relevant emotional tokens are used during inference, predictions could remain unstable when irrelevant context or background redundancy dominates. Fig. 1 provides a conceptual comparison between fine-tuning, training-free prompting, and our inference-time token refinement for MER. This leads to a fundamental question: *Is it possible to stabilize MER without retraining by explicitly steering inference toward decision-relevant emotional tokens?*

In this work, we rethink the role of decision-relevant emotional tokens in MER from a world-model perspective, motivated by the observation that emotion recognition can be inherently a problem of latent state inference rather than direct pattern matching. Rather than direct observations from multimodal inputs, we hypothesize that emotional states manifest through sparse, fine-grained cues, such as subtle facial muscle movements or contrastive turns in text, embedded within substantial irrelevant or redundant context. Accordingly, we formulate MER as inferring a latent emotional world state from multimodal observations that mix informative emotional evidence with modality-specific noise. Under this formulation, when MLLMs are used with frozen parameters, directly disentangling emotional signals from noise by modifying model representations could be infeasible. Robust inference thus depends on structuring how observed evidence is used, rather than learning new representations. In MLLMs, multimodal inputs are processed only through tokens, which serve as the interface for internal reasoning. From a world-model perspective, only tokens that meaningfully constrain the latent emotional state should dominate inference, while noise-dominated tokens should be suppressed. This task-induced representation

makes explicit what it means for evidence to be relevant, and motivates controlling token participation and influence during inference as a principled mechanism for robust MER, forming the basis of our method.

Driven by the world-model assumption that robust emotion recognition requires prioritizing evidence that constrains the latent state, we propose **WETR** (World-Model Inspired Emotion-aware Token Refinement), a training-free, plug-and-play inference regulator for frozen MLLMs. WETR consists of two complementary components that bridge raw observations to the latent emotional state: (1) **Noise-suppressed Token Selection (NTS)** acts as an **observation filter**, isolating a compact set of emotion-relevant tokens from high-dimensional input; this reflects the world-model premise that only sparse, task-relevant observations meaningfully reduce uncertainty about the latent emotional state. (2) **State-strengthened Token Reweighting (STR)** serves as a **likelihood modulator**, adjusting token influence during multimodal aggregation to emphasize decision-relevant emotional tokens that provide the strongest constraints on the latent state while down-weighting uninformative tokens. By iteratively refining token these components at sparse trigger layers, WETR encourages inference to focus on state-constraining emotional evidence while suppressing noise, yielding more stable predictions and better token-level interpretability without parameter updates. **Our main contributions are summarized as follows:**

- We propose WETR, a training-free, plug-and-play inference-time regulator for MLLMs, inspired by a world-model perspective to explicitly refine decision-relevant emotional token usage for robust MER under noisy context.
- We introduce Noise-suppressed Token Selection (NTS), which selects a compact emotion-relevant set to suppress intra-modal redundancy while preserving decision-relevant emotional cues.
- We design State-strengthened Token Reweighting (STR), which reweights retained tokens to strengthen complementary cues and reduce the influence of weakly informative tokens.
- We validate WETR on MAFW, CH-SIMS, and MER23, demonstrating consistent improvements in accuracy and stability.

2. Related Work

2.1. MLLMs-based MER

Large Multimodal Language Models (MLLMs) have advanced MER by enabling unified, instruction-following reasoning across modalities via shared language representa-

tions (Yang et al., 2025; Liang et al., 2025). Emotion-LLaMA (Cheng et al., 2024) designed some emotion-specific encoders and employed instruction tuning to align and fuse multimodal features, significantly enhancing the ability of emotion recognition and reasoning. Affect-GPT (Lian et al., 2025) constructed the largest fine-grained descriptive emotion dataset MER-Caption and a unified benchmark MER-UniBench specifically designed for evaluating the MER tasks of MLLMs. R1-Omni (Zhao et al., 2025) applied Reinforcement Learning with Verifiable Reward (RLVR) to multimodal emotion understanding tasks for the first time, significantly enhancing the generalization ability, reasoning ability and interpretability of MLLMs in out-of-distribution scenarios. Despite strong understanding, these methods require fine-tuning, leading to high training costs. We aim to enhance the emotional recognition capabilities of MLLMs during inference, without relying on extra training, making emotional understanding more scalable.

2.2. Training-Free MLLMs

Due to the high training costs associated with MLLMs, training-free methods have gained considerable attention as a fresh perspective (Bi et al., 2025; Kim et al., 2024a; Chen et al., 2024a; Kim et al., 2024b), particularly in tasks such as inference acceleration and mitigating MLLM hallucinations. Typical training-free methods such as text prompt engineering to guide model behavior and improve outcomes without additional training, token merging to enhance processing efficiency by simplifying input data, and visual augmentation alongside multi-step inference to boost performance through visual enhancements and iterative processing. In emotion-related tasks, SEPM (Fang et al., 2025) uses confidence-guided coarse-to-fine prompting with a Focus-on-Emotion query to drop emotion-irrelevant visual tokens, improving recognition. However, most training-free strategies act on the input prompting or compress tokens generically, offering limited control over token usage. In contrast, we target inference-time token usage directly by refining both token admission and reweighting to stabilize MER under noisy context.

3. The Proposed Method

3.1. Problem Definition

Given a video-text pair (V, T) where V is a sequence of frames and T is the associated text, our goal is to predict an emotion state $\hat{y}$ via frozen MLLMs. A frozen MLLM serializes (V, T) into a unified token sequence $\mathbf{x} = [x_1, \dots, x_L]$, where x_i is the i -th token and L is the sequence length. We use index sets to distinguish token types: $\mathcal{I}_v \subset \{1, \dots, L\}$ for visual tokens and $\mathcal{I}_t \subset \{1, \dots, L\}$ for text tokens. To avoid template tokens contaminating statistics, we further split text tokens into content tokens $\mathcal{I}_{con} \subseteq \mathcal{I}_t$ and instruc-

tion tokens $\mathcal{I}_{inst} \subseteq \mathcal{I}_t$, where $\mathcal{I}_{inst}$ is always preserved to keep prompting stable and outputs parsable.

3.2. Token Usage under World-Model View

World-Model Formulation for MER To investigate how decision-relevant emotional tokens are used during inference, we adopt a world-model view of MER. We treat emotion as a latent emotional world state z_e that gives rise to multimodal observations, while each modality also contains noise factors that are irrelevant to emotion. Concretely, we abstract a video-text pair (V, T) as

$$V = g_v(z_e, \varepsilon_v), \quad T = g_t(z_e, \varepsilon_t), \quad (1)$$

where ε_v and ε_t capture modality-specific noise (e.g., background redundant, irrelevant context), and $g_v(\cdot), g_t(\cdot)$ are generative mappings. This formulation does not assume access to g_v, g_t ; it is used to clarify the inference goal: robust MER should reduce uncertainty about z_e using sparse emotion-relevant cues while resisting ε -dominated redundancy. When MLLMs are used with frozen parameters, directly learning new representations to disentangle z_e from ε is infeasible. Therefore, robustness must come from *structuring how observed evidence is used* during inference, rather than modifying representations via training. Since MLLMs process multimodal inputs purely through tokens, this world-model view points to a principled and controllable lever: *token usage*.

Token Usage as the Controllable Inference Lever Let $\mathbf{x} = [x_1, \dots, x_L]$ be the unified token sequence serialized from (V, T) . Under frozen parameters, our intervention is restricted to inference-time token participation and influence. We therefore introduce a token-usage weight vector $w = \{w_i\}_{i=1}^L$, where w_i controls how strongly token x_i contributes to the model’s internal aggregation for inferring z_e . We factorize token usage as

$$w_i = m_i \gamma_i, \quad (2)$$

where $m_i \in \{0, 1\}$ is a **selection factor** that decides whether token i is admitted as evidence, and $\gamma_i > 0$ is a **reweighting factor** that modulates the influence of admitted tokens. We denote the admitted set as $\mathcal{K} = \{i \mid m_i = 1\}$.

Our WETR. We therefore propose **WETR** (World-Model inspired Emotion-aware Token Refinement), a plug-and-play inference-time regulator that sparsely updates the two factors of w_i at a set of trigger layers. Specifically, at each trigger layer (as shown in Fig. 2), **NTS** updates m_i and acts as an **observation filter**, selecting a compact set of emotion-relevant tokens to suppress intra-modality redundancy; **STR** updates γ_i and acts as a **likelihood modulator**, re-scaling the effective likelihood contribution of retained tokens to down-weight predictable, weakly emotion-relevant

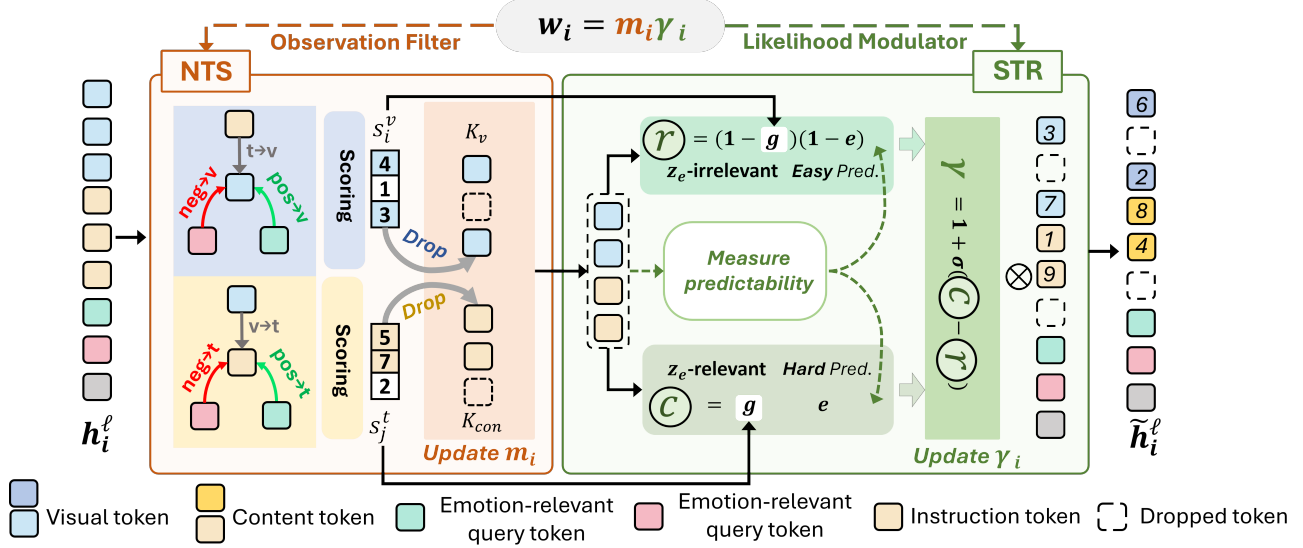


Figure 2. Our WETR at trigger layer ℓ . Given the token hidden states $\{h_i^\ell\}$, WETR refines token usage $w_i = m_i \gamma_i$ via two modules. **NTS** (observation filter) scores visual/content tokens using contrastive emotion queries and updates the selection factor m_i to keep a compact emotion-relevant set $\mathcal{K}$. **STR** (likelihood modulator) measures cross-modal predictability within $\mathcal{K}$, then updates the reweighting factor γ_i to down-weight predictable, weakly emotion-relevant overlap and up-weight hard-to-predict emotion cues. The refined hidden states $\tilde{h}_i^\ell = \gamma_i h_i^\ell$ are passed to subsequent layers, while non-trigger layers remain unchanged.

overlap and up-weight emotion-relevant residual cues that provide larger incremental constraints on z_e . We define the trigger set as $\mathcal{L}_{reg} = \{\ell \mid \ell \bmod n_r = 0\}$ and leave all non-trigger layers unchanged. The final prediction $\hat{y}$ is produced by standard decoding on the refined hidden states. Overall, this sparse refinement progressively shifts inference away from ε -dominated noisy tokens toward tokens that better constrain z_e , improving stability without any parameter updates.

3.3. NTS: Noise-suppressed Token Selection

Goal. As the **observation filter** in WETR, NTS updates the **selection factor** m_i in $w_i = m_i \gamma_i$ to control token admission under frozen parameters. It constructs a compact emotion-relevant evidence set $\mathcal{K}$ that suppresses intra-modality redundancy (tokens largely explained by ε) while retaining sparse cues that are most informative for inferring the latent emotion state z_e . Formally, the desired utility of admitting token x_i is its conditional information gain,

$$\Delta_i^* = I(z_e; x_i \mid \mathbf{x}_{-i}), \quad (3)$$

which measures how much x_i reduces uncertainty about z_e beyond what is already explained by the remaining tokens. Since Δ_i^* is not directly accessible in frozen MLLMs, NTS estimates a computable proxy $\hat{\Delta}_i$ during inference and admits the top-scoring tokens into $\mathcal{K}$.

Contrastive-query utility proxy. To obtain a computable proxy for the conditional gain in Eq. (3), we score each

token by *contrastive querying*. We insert two short instruction spans: an **emotion-relevant** query Q_{pos} that explicitly seeks decision-relevant emotional cues, and an **emotion-irrelevant** query Q_{neg} that can be answered without emotion, thus reflecting generic context usage. Let their spans be $\mathcal{I}_{\text{pos}}, \mathcal{I}_{\text{neg}} \subseteq \mathcal{I}_{\text{inst}}$. At trigger layer ℓ , with head-aggregated attention $\mathbf{A}^{(\ell)}$, denote by $a_{\mathcal{I} \rightarrow i}^{(\ell)}$ the average attention mass from span $\mathcal{I}$ to token i . We first compute a contrastive utility

$$d_i = \text{Norm} \left(a_{\mathcal{I}_{\text{pos}} \rightarrow i}^{(\ell)} - a_{\mathcal{I}_{\text{neg}} \rightarrow i}^{(\ell)} \right). \quad (4)$$

We use the difference rather than $a_{\mathcal{I}_{\text{pos}} \rightarrow i}^{(\ell)}$ alone to cancel query-shared, ε -driven attention and prompt-induced global bias, so d_i better reflects the incremental evidence of x_i for constraining z_e in Eq. (3). As a result, tokens mainly explained by irrelevant context tend to receive similar attention from both queries and are suppressed after subtraction, while emotion-informative tokens remain salient. To reduce modality bias, we add a weak cross-modal support term and define the final selection score:

$$\hat{\Delta}_i = s_i = d_i + \begin{cases} \text{Norm} \left(\frac{1}{|\mathcal{I}_{\text{con}}|} \sum_{q \in \mathcal{I}_{\text{con}}} \mathbf{A}_{q,i}^{(\ell)} \right), & i \in \mathcal{I}_v, \\ \text{Norm} \left(\frac{1}{|\mathcal{I}_v|} \sum_{q \in \mathcal{I}_v} \mathbf{A}_{q,i}^{(\ell)} \right), & i \in \mathcal{I}_{\text{con}}. \end{cases} \quad (5)$$

We use s_i as the proxy utility $\hat{\Delta}_i$ for ranking tokens toward the information-gain objective Δ_i^* .

Top- p emotion-relevant set construction. Using the final scores s_i , NTS admits only the highest-utility tokens to form a compact emotion-relevant set. To prevent a single

modality from dominating, we apply top- p filtering within each modality: we keep a ratio p_v of visual tokens and a ratio p_t of text-content tokens. Let $\text{TopP}(\cdot, p)$ return the indices of the top- p fraction ranked by scores; we obtain

$$\mathcal{K}_v = \text{TopP}(\{s_i\}_{i \in \mathcal{I}_v}, p_v), \mathcal{K}_{\text{con}} = \text{TopP}(\{s_j\}_{j \in \mathcal{I}_{\text{con}}}, p_t). \quad (6)$$

The final emotion-relevant set is $\mathcal{K} = \mathcal{K}_v \cup \mathcal{K}_{\text{con}} \cup \mathcal{I}_{\text{inst}}$, where $\mathcal{I}_{\text{inst}}$ is always preserved for prompt stability and parsable outputs. Accordingly, the selection factor is

$$m_i = \mathbb{I}[i \in \mathcal{K}], \quad (7)$$

where $\mathbb{I}[\cdot]$ is the indicator function.

3.4. STR: State-strengthened Token Reweighting

Goal. As the **likelihood modulator** in WETR, STR updates the **reweighting factor** γ_i in $w_i = m_i \gamma_i$ to control how strongly retained tokens contribute to aggregating evidence for the latent emotion state z_e . Under the world-model view, each retained token can be regarded as providing a likelihood term $\log p(x_i | z_e)$; reweighting γ_i corresponds to modulating its effective likelihood contribution, i.e.,

$$\log \tilde{p}(z_e | \mathbf{x}; w) \propto \log p(z_e) + \sum_{i \in \mathcal{K}} (m_i \gamma_i) \log p(x_i | z_e). \quad (8)$$

Since $\log p(x_i | z_e)$ is not directly accessible in frozen MLLMs, STR aims to redistribute γ_i within $\mathcal{K}$ so that tokens with higher incremental utility for constraining z_e receive larger weights, while easy-to-match, ε -driven overlap with low marginal gain is down-weighted. For stability, STR is gated by the NTS score and preserves a small set of high-confidence anchors.

Let $\mathbf{H} \in \mathbb{R}^{L \times d}$ be the hidden states at trigger layer ℓ (token i : $\mathbf{h}_i$), and let $\mathbf{q}_i, \mathbf{k}_i$ denote its query/key vectors from the frozen attention projections.

Identifying replaceable overlap vs. complementary residual evidence. To realize the likelihood modulation in Eq. (8) without training, STR needs a computable cue to redistribute γ_i within $\mathcal{K}$. We distinguish (i) *replaceable overlap* that is largely predictable from the other modality under the current alignment, from (ii) *complementary residual evidence* that is hard to predict and thus can provide larger incremental constraints on z_e . This does not dismiss agreement: if x_i is already well explained by $\mathbf{x}_{-i}$, its conditional gain $I(z_e; x_i | \mathbf{x}_{-i})$ is typically small, so increasing its likelihood weight yields limited posterior tightening. Accordingly, STR mainly down-weights predictable but weakly emotion-relevant tokens, keeps emotion-relevant agreement largely unchanged, and up-weights emotion-relevant residual cues.

We use cross-modal predictability as a lightweight indicator of replaceability. Under the query-induced alignment at

layer ℓ , tokens whose keys are well reconstructable from the other modality are treated as overlap, while poorly reconstructable ones are treated as residual evidence. For $i \in \mathcal{K}_v$, we compute

$$\alpha_{ij} = \text{Softmax}_{j \in \mathcal{K}_{\text{con}}} \left(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d}} \right), \quad (9)$$

and measure predictability by the reconstruction error $e_i^v = \left\| \mathbf{k}_i - \sum_{j \in \mathcal{K}_{\text{con}}} \alpha_{ij} \mathbf{k}_j \right\|_2$. We define e_j^t symmetrically (reconstructing from $\mathcal{K}_v$) and normalize all errors to $\bar{e}_i \in [0, 1]$. Predictability alone does not indicate emotion relevance, so we gate it by the NTS score s_i : $g_i = \varphi(\text{Norm}(s_i)) \in [0, 1]$, where $\varphi(\cdot)$ is a sigmoid gate. We then define

$$r_i = (1 - g_i)(1 - \bar{e}_i), \quad c_i = g_i \bar{e}_i, \quad (10)$$

where r_i highlights predictable but weakly emotion-relevant overlap, while c_i highlights emotion-relevant yet hard-to-predict residual evidence. Notably, emotion-relevant agreement corresponds to high g_i and low $\bar{e}_i$, yielding small r_i and c_i , so it is not down-weighted ($\gamma_i \approx 1$).

Reweighting update with anchor protection. Based on the overlap/residual scores (r_i, c_i), STR updates γ_i to implement the likelihood modulation in Eq. (8): it down-weights predictable but weakly emotion-relevant overlap (large r_i) and up-weights emotion-relevant residual evidence (large c_i). We use a signed migration bias

$$b_i = c_i - r_i, \quad \gamma_i = 1 + \delta b_i, \quad (11)$$

where $\delta > 0$ controls the intervention strength. To avoid over-suppressing reliable evidence, we protect a small anchor set $\mathcal{A}$ selected by NTS scores (top- p_v^{anc} within $\mathcal{K}_v$ and top- p_t^{anc} within $\mathcal{K}_{\text{con}}$). For anchors, we enforce a lower bound by clamping $\gamma_i \leftarrow \max(\gamma_i, \gamma_{\text{anc}})$, where $\gamma_{\text{anc}} \geq 1$. Finally, we apply token-wise scaling $\tilde{\mathbf{h}}_i = \gamma_i \mathbf{h}_i$ so subsequent layers attend on $\tilde{\mathbf{h}}_i$, realizing the refined usage $w_i = m_i \gamma_i$.

4. Experiment

We design experiments to answer the following questions:

RQ1: Does WETR consistently improve MER across frozen MLLMs under various datasets? (Sec. 4.3)

RQ2: Can NTS select compact emotion-relevant evidence by suppressing ε -dominated tokens? (Sec. 4.4)

RQ3: Can STR reweight retained tokens to strengthen state-informative emotional cues? (Sec. 4.4)

RQ4: Under noise amplification ($\varepsilon \uparrow$), can WETR maintain z_e -constrained token usage? (Sec. 4.4)

Table 1. Comparison with state-of-the-art on the MAFW, CH-SIMS and MER23 database. The best results are in **bold**.

Methods	Speed	MAFW dataset			SIMS dataset			MER23 dataset		
		WAR	UAR	F1	ACC-2	ACC-3	F1	WAR	UAR	F1
Qwen2-VL-7B (Wang et al., 2024)	5.4010s	43.58	36.82	47.68	76.55	48.49	75.68	57.66	54.05	60.78
Janus-Pro-1B (Chen et al., 2025a)	0.5730s	39.36	24.28	32.30	61.08	60.83	61.83	44.04	43.87	44.34
InterVL2.5-2B (Chen et al., 2025b)	0.8791s	52.76	40.66	53.75	65.98	54.05	74.44	44.28	41.95	46.02
InterVL2.5-4B (Chen et al., 2025b)	1.1857s	49.55	39.67	50.51	86.08	66.30	86.20	63.26	55.32	62.46
InterVL3-8B (Zhu et al., 2025)	1.0287s	53.54	42.28	52.47	86.86	62.14	87.00	56.69	52.80	56.18
HumanOmni-0.5B (Zhao et al., 2025)	—	20.18	13.52	—	—	—	—	—	—	—
EMER-SFT (Zhao et al., 2025)	—	38.39	28.02	—	—	—	—	—	—	—
PandaGPT (Su et al., 2023)	—	—	—	—	62.37	—	62.93	39.13	—	—
Emotion-LLaMA (Cheng et al., 2024)	—	—	—	—	78.32	—	78.61	59.38	—	—
MoSE (Han et al., 2025)	2.1169s	33.66	26.85	33.04	69.07	52.74	62.13	51.84	49.85	57.24
<i>InterVL3.5-4B (Wang et al., 2025)</i>										
Zero-shot	1.4583s	57.76	45.46	54.73	86.60	65.21	86.60	60.83	59.84	62.03
Zero-shot-CoT (Kojima et al., 2022)	10.2317s	57.62	43.19	55.40	86.51	64.30	86.74	57.42	56.54	58.33
SparseVLM (Zhang et al., 2024c)	1.9884s	55.36	42.67	53.27	85.15	62.36	85.50	61.80	58.79	62.83
SEPM (Fang et al., 2025)	6.0915s	56.45	44.12	54.11	84.46	57.91	84.72	59.12	58.19	60.73
Ours	1.3487s	60.96	46.95	57.08	87.63	69.15	87.68	64.72	59.86	65.17
<i>InterVL3.5-8B (Wang et al., 2025)</i>										
Zero-shot	1.7089s	55.08	44.21	54.19	86.34	62.36	86.49	60.34	57.36	60.57
Zero-shot-CoT (Kojima et al., 2022)	11.0172s	55.20	45.20	55.19	86.86	67.18	87.00	57.66	54.01	58.19
SparseVLM (Zhang et al., 2024c)	1.7310s	55.32	44.39	55.01	86.18	65.21	86.80	62.53	58.14	63.29
SEPM (Fang et al., 2025)	4.8369s	56.15	45.37	56.83	87.25	68.88	87.18	63.26	57.48	62.73
Ours	1.6573s	59.64	46.30	57.57	89.18	70.90	89.16	68.13	63.24	68.48
<i>Qwen3-VL-4B (Bai et al., 2025)</i>										
Zero-shot	2.2669s	50.08	42.03	50.21	85.31	68.71	85.22	56.93	56.19	58.22
Zero-shot-CoT (Kojima et al., 2022)	8.1134s	49.66	40.01	52.38	86.02	70.51	85.98	57.66	56.66	59.05
SparseVLM (Zhang et al., 2024c)	3.211s	50.39	41.15	50.30	86.68	74.84	86.71	57.91	57.63	59.12
SEPM (Fang et al., 2025)	1.4923s	51.66	42.56	51.15	86.93	70.87	86.39	58.65	58.50	60.25
Ours	2.5953s	53.29	44.02	53.91	87.63	76.15	87.59	61.17	61.56	62.46
<i>Qwen3-VL-8B (Bai et al., 2025)</i>										
Zero-shot	2.6964s	52.78	43.35	52.50	85.57	65.89	85.72	65.94	63.89	66.16
Zero-shot-CoT (Kojima et al., 2022)	14.5517s	51.99	42.02	51.29	85.76	71.65	85.54	66.91	64.57	66.32
SparseVLM (Zhang et al., 2024c)	6.390s	52.60	42.36	52.33	87.31	73.09	87.02	66.91	64.57	66.32
SEPM (Fang et al., 2025)	2.103s	54.27	44.47	53.93	86.42	69.58	86.02	66.78	65.21	66.51
Ours	3.0392s	56.79	46.93	56.83	87.89	76.43	88.09	68.37	66.29	68.70

4.1. Datasets

We conducted experiments on three widely used multimodal emotion datasets: MAFW (Liu et al., 2022), CH-SIMS (Yu et al., 2020) and MER23 (Lian et al., 2023). **MAFW** contains 10,045 video clips with 11 basic and 32 compound emotion labels; we follow the standard five-fold cross-validation protocol for comparisons. **CH-SIMS** is a Chinese dataset with 2,281 clips annotated by a continuous sentiment score in $[-1, 1]$ with modality-level annotations. **MER23** provides short video clips with aligned audio and text for 6 discrete emotion classification (neutral, anger, happiness, sadness, worry, surprise).

4.2. Experimental Setting

Following prior work (Liu et al., 2022), we adopt standard evaluation metrics for each dataset. For MAFW and MER23, we report Unweighted Average Recall (UAR), Weighted Average Recall (WAR), and macro-averaged F1-score. For CH-SIMS, we report 2-class accuracy (Acc-2), 3-class ac-

curacy (Acc-3), and macro-averaged F1-score. All experiments were conducted on a Linux workstation with a single NVIDIA A100 Tensor Core GPU, and our method was implemented in PyTorch. We evaluate WETR on representative MLLMs (InternVL3.5-4B/8B and Qwen3-VL-4B/8B). For all experiments, we perform inference with publicly released pre-trained checkpoints without extra training.

4.3. Comparison with State-of-the-Art Methods (RQ1)

We evaluate WETR on three MER benchmarks (MAFW, CH-SIMS, MER23) with four frozen MLLM backbones (InternVL3.5-4B/8B and Qwen2.5-4B/8B), and compare against Zero-shot, Zero-shot-CoT, and prior training-free baselines (as shown in Tab. 1). For Zero-shot-CoT, we append “let’s think step by step” as a prompt enhancement. Across all datasets and backbones, WETR consistently improves MER without changing model parameters or architecture, demonstrating strong cross-model generalization. In particular, WETR brings clear gains over Zero-shot on both InternVL3.5-8B (+4.56% WAR on MAFW) and Qwen2.5-

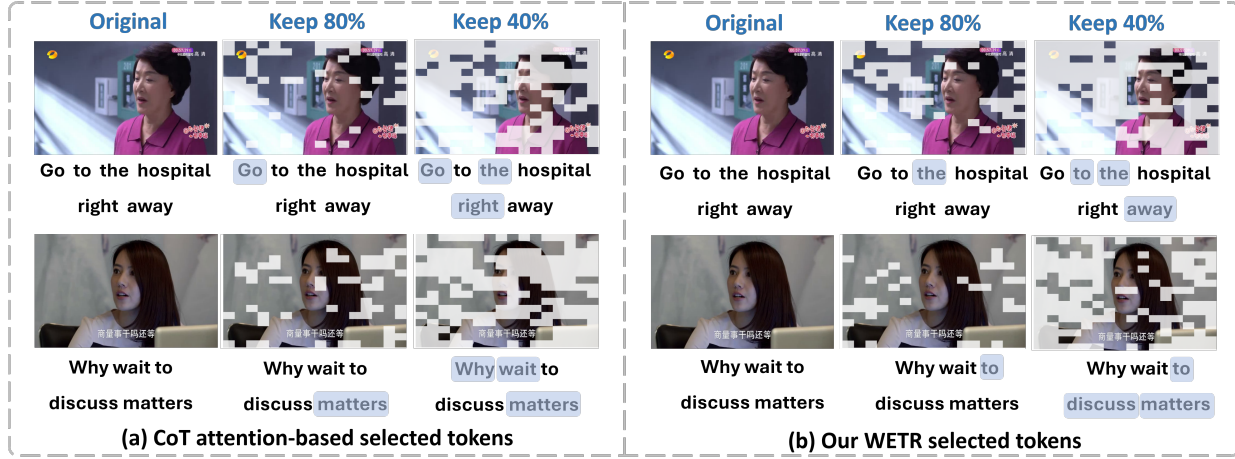


Figure 3. Visualization of selected tokens for different keep ratios. The occluded part represents dropped tokens.

8B (+4.01% WAR on MAFW), and yields the largest improvement on MER23 (+7.79% WAR on InternVL3.5-8B). In terms of efficiency, WETR maintains comparable latency to Zero-shot, whereas Zero-shot-CoT is consistently much slower (about $4 \times - 7 \times$ across backbones). Overall, WETR offers a favorable accuracy-efficiency trade-off and consistently improves MER across datasets and frozen MLLMs (answering **RQ1**).

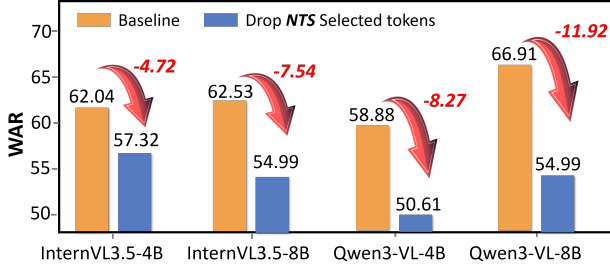


Figure 4. Effect of removing NTS-selected tokens across frozen MLLMs. Dropping the tokens selected by NTS causes a sharp WAR decrease on all backbones, indicating that NTS captures decision-critical emotion evidence.

4.4. Ablation Studies

Comparison of Token Selection Strategies. We study how the NTS scoring proxy affects evidence selection on MER23 with InternVL3.5-8B fully frozen. All variants keep the same token ratio and differ only in the scoring function (Tab. 2). Moving from Random selection to attention-based scoring consistently improves performance, indicating that NTS can identify a compact, informative evidence set rather than arbitrary tokens. Among these proxies, the proposed contrastive score is the most effective: it achieves the best WAR/UAR/F1 and improves UAR by +6.17% over Random (and +1.76% over using Q_{pos} alone), showing that contrasting emotion-seeking and noise-seeking signals better suppresses ϵ -dominated redundancy and selects more emotion-relevant tokens (answering **RQ2**).

Key Component Ablation Analysis. We ablate WETR on MER23 to isolate the contributions of NTS and STR (Tab. 3). Both modules individually improve the frozen baseline, with NTS providing the larger boost and STR also yielding consistent gains. Combining both achieves the best WAR/UAR/F1, indicating that the two modules are complementary: NTS first selects a compact, emotion-relevant evidence set to suppress ϵ -dominated redundancy, and STR then reweights the retained tokens to further strengthen state-informative cues (supporting **RQ2-RQ3**).

Table 2. The effect of different proxy token-usage scoring strategies in NTS on the MER23 database. The best results are in **bold**.

Selection strategies	WAR	UAR	F1
Random	64.48	57.07	64.56
Query-related	66.67	60.07	66.76
Q_{pos} -related	67.40	61.48	67.58
Contrastive-query-related	68.13	63.24	68.48

Table 3. The effect of different modules on model performance on the MER23 database. The best results are in **bold**.

Baseline	NTS	STR	WAR	UAR	F1
✓			62.53	58.31	63.26
✓	✓		67.64	61.68	67.84
✓		✓	65.94	60.03	66.22
✓	✓	✓	68.13	63.24	68.48

Visualization and Deletion Test of Selected Tokens.

Fig. 3 qualitatively shows what NTS removes under different keep ratios: it mainly drops ϵ -dominated visual background/redundant regions and weakly-informative text fragments, while preserving compact facial areas and decisive emotion triggers. To provide causal evidence, we further remove the NTS-selected tokens and keep all others unchanged. As shown in Fig. 4, this deletion causes a pronounced WAR degradation across all frozen backbones, indicating that the selected set is decision-critical. Together, these results confirm that NTS constructs a compact,

emotion-relevant evidence set by suppressing ε -dominated tokens (answering **RQ2**).

Visualization of Inference-time Token Influence Redistribution. Fig. 5 visualizes attention maps on MER23 to assess whether STR can redistribute token influence toward z_e -constraining cues. Compared with Base, Base+STR more often concentrates attention on facial regions, indicating that reweighting alone already provides a moderate sharpening effect. Meanwhile, without an explicit evidence filter, its attention may still be partially drawn to visually salient yet emotion-weak regions in some cases. By first filtering noisy tokens, NTS yields a cleaner evidence set and makes STR’s reweighting more consistent. Consequently, WETR (NTS+STR) exhibits the most concentrated and stable attention on faces, supporting effective inference-time influence redistribution (answering **RQ3**).

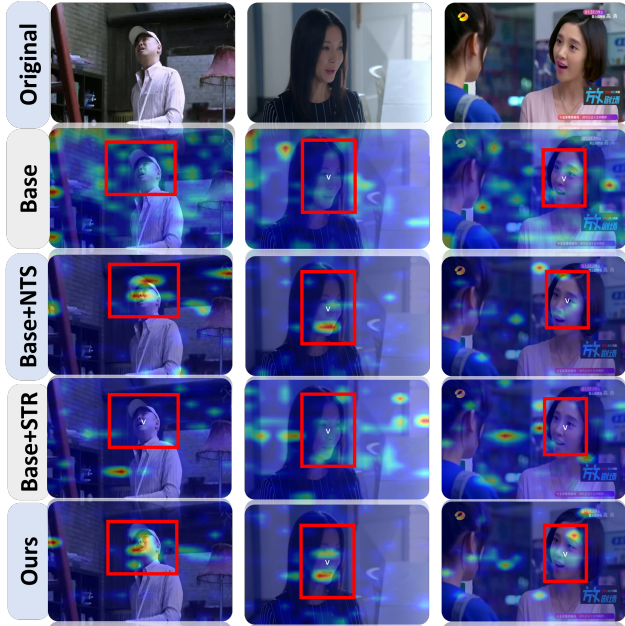


Figure 5. Attention heatmaps on MER23. We compare Base, Base+NTS, Base+STR, and WETR (Ours); red boxes mark facial regions. NTS sharpens attention by filtering evidence, while STR further concentrates attention on emotion-relevant facial cues; combining both yields the most stable and focused attention.

Robustness Analysis under Noise Amplification. To test robustness, we increase ε to inject more emotion-irrelevant redundancy into both video and text with frozen InternVL3.5-8B. We report F1 and Emotion Attention Mass (EMass), which measures how much attention is allocated to a shared evidence set $\mathcal{K}$ under an emotion-seeking query versus a noise-seeking query (Fig. 6). As ε increases, WETR degrades more slowly than the baseline and consistently preserves higher EMass, indicating that it better maintains z_e -constrained token usage under amplified nuisance. At the highest noise level ($\varepsilon=7$), WETR improves F1 over the baseline by 9.18%. Overall, WETR remains robust under

controlled nuisance amplification (answering **RQ4**).

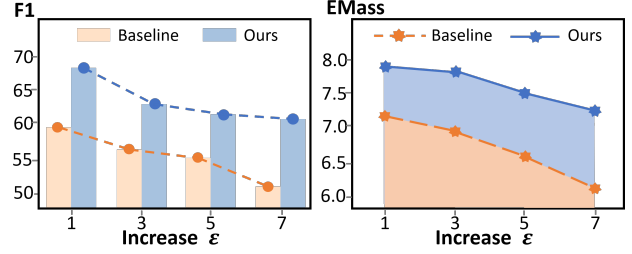


Figure 6. Robustness under noise amplification on the MER23 database. Left: F1 as ε increases. Right: Emotion Attention Mass (EMass), measuring the attention mass captured by a shared evidence set under emotion-seeking vs noise-seeking queries.

Parameter Sensitivity Analysis. We study the sensitivity of WETR on MER23 (InternVL3.5-8B frozen) w.r.t. the trigger interval n_r , intervention strength δ , and modality keep ratios (p_v, p_t) (Tab. 4, Fig. 7). As shown in Tab. 4, both overly frequent and overly sparse refinements are suboptimal; a moderate interval performs best, and we set $n_r=9$. Similarly, δ is robust within a reasonable range, and $\delta=4$ gives the best overall results. Fig. 7 further shows that performance peaks at compact evidence sizes: keeping a small fraction of visual tokens ($p_v=0.03$) and a moderate fraction of text tokens ($p_t=0.2$) yields the best trade-off, while retaining too many tokens introduces ε -dominated redundancy and retaining too few removes useful cues. We use $n_r=9$, $\delta=4$, $p_v=0.03$, and $p_t=0.2$ for all experiments.

Table 4. The effect of different interval layers n_r and intervention strength δ on the MER23 database. The best results are in **bold**.

n_r	WAR	UAR	F1	δ	WAR	UAR	F1
2	66.67	61.75	67.11	1	67.40	61.60	67.50
3	67.40	59.81	67.17	2	67.15	61.98	67.41
6	66.91	60.95	67.10	3	67.64	62.98	68.04
9	68.13	63.24	68.48	4	68.13	63.24	68.48
18	66.42	61.69	66.87	5	68.07	62.64	68.33

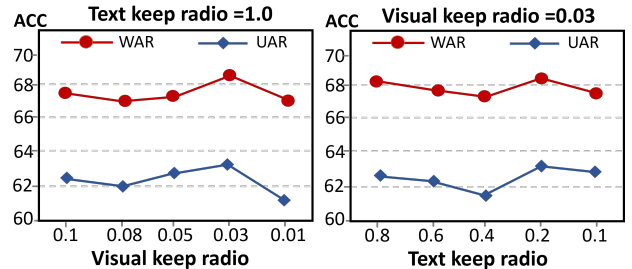


Figure 7. The effect of different modality keep ratios on the MER23 database.

Error Propagation Analysis. We study how irreversible token dropping affects inference on MER23 with InternVL3.5-8B fully frozen. Specifically, we inject controlled post-NTS selection noise with ratio r and compare the frozen baseline, NTS-only, and full NTS+STR (Tab. 5).

Table 5. Error propagation analysis under controlled post-NTS noise injection on the MER23 database. Values are F1, with Δ F1 from the clean setting shown in parentheses.

Method	$r=0\%$	$r=5\%$	$r=10\%$	$r=20\%$	$r=30\%$
Baseline	60.57 (0.00)	59.12 (-1.45)	59.01 (-1.56)	56.02 (-4.55)	53.65 (-6.92)
NTS-only	67.84 (0.00)	67.62 (-0.22)	67.23 (-0.61)	66.24 (-1.60)	65.14 (-2.70)
NTS+STR	68.48 (0.00)	68.26 (-0.18)	68.04 (-0.44)	67.46 (-1.02)	66.72 (-1.76)

As r increases, performance degrades for all methods, but the full model degrades much more slowly than both the frozen baseline and NTS-only, indicating that STR effectively mitigates error accumulation after hard token selection. In particular, at $r=30\%$, the F1 drop of NTS+STR is only 1.76%, compared with 2.70% for NTS-only and 6.92% for the frozen baseline. We further examine the role of anchor protection in STR. As shown in Tab. 6, the full model consistently outperforms both NTS-only and the variant without anchor protection, suggesting that anchor protection further stabilizes inference after token filtering. Overall, these results show that although NTS is irreversible, its error propagation remains well controlled in WETR design.

Table 6. Effect of anchor protection on the MER23 database.

Setting	WAR	UAR	F1
NTS only	67.64	61.68	67.84
NTS + STR w/o anchor protection	67.84	62.75	68.07
NTS + STR (full)	68.13	63.24	68.48

Prompt Robustness of q_{pos} and q_{neg} . We evaluate three prompt variants for each query strategy on MER23 with InternVL3.5-8B fully frozen (Tab. 7). Using either Q_{pos} or Q_{neg} alone yields lower average performance and larger variance across prompt variants, indicating that single-query scoring is more sensitive to wording changes. By contrast, the proposed contrastive strategy achieves the best average F1 and the smallest standard deviation. Specifically, $Q_{pos} - Q_{neg}$ reaches 68.17% F1, outperforming Q_{pos} only and Q_{neg} only by 1.18% and 2.18%, respectively, while reducing the standard deviation to 0.40. These results suggest that the contrastive formulation provides a more stable estimate of emotion-relevant token utility.

Table 7. Performance and variance of different query scoring strategies under prompt variants on the MER23 database.

Strategy	V1	V2	V3	Mean $\pm$ Std
Q_{pos} only	67.58	67.15	66.24	66.99 $\pm$ 0.68
Q_{neg} only	66.76	66.72	64.49	65.99 $\pm$ 1.30
$Q_{pos} - Q_{neg}$	68.48	67.71	68.32	68.17$\pm$0.40

Token Decision Contribution Shift Analysis. We study how STR affects the decision contribution of retained tokens on MER23 with InternVL3.5-8B fully frozen. Specifically, we group retained tokens by their reweighting factor γ_i into three categories: downweighted (DW), neutral (NEU), and upweighted (UW), and compare their average contribution before and after STR, where contribution is mea-

sured by the drop in the target prediction score when the token is removed once (Tab. 8). Tokens with larger positive reweighting factors show a clear contribution increase after STR, whereas downweighted tokens exhibit a substantial decrease, indicating that STR does not simply rescale all retained tokens uniformly. In particular, the contribution of UW tokens increases by 0.0225, while that of DW tokens decreases by 0.0196; by contrast, NEU tokens remain nearly unchanged. These results suggest that STR selectively redistributes token influence by strengthening more state-informative evidence and suppressing weaker or more replaceable cues, supporting its role as an inference-time contribution modulator (answering **RQ3**).

Table 8. Token decision contribution shift before and after STR on the MER23 database.

Group	Avg. γ_i	Before STR	After STR	Change
DW	0.000	0.0301	0.0105	-0.0196
NEU	1.000	0.0131	0.0139	+0.0008
UW	2.737	0.0381	0.0606	+0.0225

5. Conclusion

We propose **WETR**, a training-free, plug-and-play inference-time regulator for multimodal emotion recognition with frozen MLLMs. Motivated by a world-model perspective, WETR improves reliability by making token usage explicit and refining it during inference, reducing the dominance of redundant or irrelevant context while strengthening decision-relevant emotional evidence that better constrains the latent emotion state z_e . WETR performs sparse interventions at a few trigger layers via a lightweight selection-reweighting loop: **NTS** acts as an observation filter to admit a compact set of emotion-relevant tokens, and **STR** serves as a likelihood modulator to redistribute token influence using state-gated cross-modal predictability with anchor protection. Extensive experiments on MAFW, CH-SIMS, and MER23 across multiple frozen backbones demonstrate consistent improvements, together with more interpretable token-level evidence usage.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Bai, S., Cai, Y., Chen, R., Chen, K., Chen, X., Cheng, Z., Deng, L., Ding, W., Gao, C., Ge, C., Ge, W., Guo, Z., Huang, Q., Huang, J., Huang, F., Hui, B., Jiang, S., Li, Z., Li, M., Li, M., Li, K., Lin, Z., Lin, J., Liu, X., Liu, J., Liu, C., Liu, Y., Liu, D., Liu, S., Lu, D., Luo, R., Lv, C., Men, R., Meng, L., Ren, X., Ren, X., Song, S., Sun, Y., Tang, J., Tu, J., Wan, J., Wang, P., Wang, P., Wang, Q., Wang, Y., Xie, T., Xu, Y., Xu, H., Xu, J., Yang, Z., Yang, M., Yang, J., Yang, A., Yu, B., Zhang, F., Zhang, H., Zhang, X., Zheng, B., Zhong, H., Zhou, J., Zhou, F., Zhou, J., Zhu, Y., and Zhu, K. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025.
- Bi, J., Wang, Y., Yan, D., Huang, W., Jin, Z., Ma, X., Hecker, A., Ye, M., Xiao, X., Schuetze, H., et al. Prism: Self-pruning intrinsic selection method for training-free multimodal data selection. *arXiv preprint arXiv:2502.12119*, 2025.
- Chen, L., Zhao, H., Liu, T., Bai, S., Lin, J., Zhou, C., and Chang, B. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In *European Conference on Computer Vision*, pp. 19–35. Springer, 2024a.
- Chen, X., Wu, Z., Liu, X., Pan, Z., Liu, W., Xie, Z., Yu, X., and Ruan, C. Janus-pro: Unified multimodal understanding and generation with data and model scaling, 2025a. URL <https://arxiv.org/abs/2501.17811>.
- Chen, Z., Wu, J., Wang, W., Su, W., Chen, G., Xing, S., Zhong, M., Zhang, Q., Zhu, X., Lu, L., et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 24185–24198, 2024b.
- Chen, Z., Wang, W., Cao, Y., Liu, Y., Gao, Z., Cui, E., Zhu, J., Ye, S., Tian, H., Liu, Z., Gu, L., Wang, X., Li, Q., Ren, Y., Chen, Z., Luo, J., Wang, J., Jiang, T., Wang, B., He, C., Shi, B., Zhang, X., Lv, H., Wang, Y., Shao, W., Chu, P., Tu, Z., He, T., Wu, Z., Deng, H., Ge, J., Chen, K., Zhang, K., Wang, L., Dou, M., Lu, L., Zhu, X., Lu, T., Lin, D., Qiao, Y., Dai, J., and Wang, W. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling, 2025b. URL <https://arxiv.org/abs/2412.05271>.
- Cheng, Z., Cheng, Z.-Q., He, J.-Y., Wang, K., Lin, Y., Lian, Z., Peng, X., and Hauptmann, A. Emotion-llama: Multimodal emotion recognition and reasoning with instruction tuning. *Advances in Neural Information Processing Systems*, 37:110805–110853, 2024.
- Fang, Y., Liang, J., Huang, W., Li, H., Su, K., and Ye, M. Catch your emotion: Sharpening emotion perception in multimodal large language models. In *Forty-second International Conference on Machine Learning*, 2025.
- Han, Z., Zhu, B., Xu, Y., Song, P., and Yang, X. Benchmarking and bridging emotion conflicts for multimodal emotion reasoning. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 5528–5537, 2025.
- Huang, W., Ye, M., Shi, Z., Wan, G., Li, H., Du, B., and Yang, Q. Federated learning for generalization, robustness, fairness: A survey and benchmark. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):9387–9406, 2024.
- Kim, J., Yeonju, K., and Ro, Y. M. What if...?: Thinking counterfactual keywords helps to mitigate hallucination in large multi-modal models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pp. 10672–10689, 2024a.
- Kim, M., Gao, S., Hsu, Y.-C., Shen, Y., and Jin, H. Token fusion: Bridging the gap between token pruning and token merging. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 1383–1392, 2024b.
- Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., and Iwasawa, Y. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35: 22199–22213, 2022.
- Li, Z., Yang, B., Liu, Q., Ma, Z., Zhang, S., Yang, J., Sun, Y., Liu, Y., and Bai, X. Monkey: Image resolution and text label are important things for large multi-modal models. In *proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 26763–26773, 2024.
- Lian, Z., Sun, H., Sun, L., Chen, K., Xu, M., Wang, K., Xu, K., He, Y., Li, Y., Zhao, J., et al. Mer 2023: Multi-label learning, modality robustness, and semi-supervised learning. In *Proceedings of the 31st ACM international conference on multimedia*, pp. 9610–9614, 2023.
- Lian, Z., Chen, H., Chen, L., Sun, H., Sun, L., Ren, Y., Cheng, Z., Liu, B., Liu, R., Peng, X., et al. Affectgpt: A new dataset, model, and benchmark for emotion understanding with multimodal large language models. *arXiv preprint arXiv:2501.16566*, 2025.

- Liang, J., Huang, W., Wan, G., Yang, Q., and Ye, M. Lorasculpt: Sculpting lora for harmonizing general and specialized knowledge in multimodal large language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 26170–26180, 2025.
- Liu, H., Li, C., Li, Y., and Lee, Y. J. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 26296–26306, 2024.
- Liu, Y., Dai, W., Feng, C., Wang, W., Yin, G., Zeng, J., and Shan, S. Mafw: A large-scale, multi-modal, compound affective database for dynamic facial expression recognition in the wild. In *Proceedings of the 30th ACM International Conference on Multimedia*, MM ’22, pp. 24–32, New York, NY, USA, 2022. Association for Computing Machinery. ISBN 9781450392037. doi: 10.1145/3503161.3548190. URL <https://doi.org/10.1145/3503161.3548190>.
- Su, Y., Lan, T., Li, H., Xu, J., Wang, Y., and Cai, D. Pandagpt: One model to instruction-follow them all. *arXiv preprint arXiv:2305.16355*, 2023.
- Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., Fan, Y., Dang, K., Du, M., Ren, X., Men, R., Liu, D., Zhou, C., Zhou, J., and Lin, J. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution, 2024. URL <https://arxiv.org/abs/2409.12191>.
- Wang, W., Gao, Z., Gu, L., Pu, H., Cui, L., Wei, X., Liu, Z., Jing, L., Ye, S., Shao, J., et al. Internvl3. 5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.
- Xie, H., Peng, C.-J., Tseng, Y.-W., Chen, H.-J., Hsu, C.-F., Shuai, H.-H., and Cheng, W.-H. Emovit: Revolutionizing emotion insights with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 26596–26605, 2024.
- Yang, Q., Ye, M., and Du, B. Emollm: Multimodal emotional understanding meets large language models. *arXiv preprint arXiv:2406.16442*, 2024.
- Yang, Q., Bai, D., Peng, Y.-X., and Wei, X. Omni-emotion: Extending video mllm with detailed face and audio modeling for multimodal emotion analysis. *arXiv preprint arXiv:2501.09502*, 2025.
- Yu, W., Xu, H., Meng, F., Zhu, Y., Ma, Y., Wu, J., Zou, J., and Yang, K. CH-SIMS: A Chinese multimodal sentiment analysis dataset with fine-grained annotation of modality. In Jurafsky, D., Chai, J., Schluter, N., and Tetreault, J. (eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 3718–3727, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.343. URL <https://aclanthology.org/2020.acl-main.343/>.
- Zhang, L., Luo, Z., Wu, S., and Nakashima, Y. Microemo: Time-sensitive multimodal emotion recognition with subtle clue dynamics in video dialogues. In *Proceedings of the 2nd International Workshop on Multimodal and Responsible Affective Computing*, pp. 110–115, 2024a.
- Zhang, Q., Wang, Z., Zhang, D., Niu, W., Caldwell, S., Gedeon, T., Liu, Y., and Qin, Z. Visual prompting in llms for enhancing emotion recognition. *arXiv preprint arXiv:2410.02244*, 2024b.
- Zhang, S., Chen, A., Sun, Y., Gu, J., Zheng, Y.-Y., Koniusz, P., Zou, K., Hengel, A. v. d., and Xue, Y. Open eyes, then reason: Fine-grained visual mathematical understanding in mllms. *arXiv preprint arXiv:2501.06430*, 2025.
- Zhang, Y., Fan, C.-K., Ma, J., Zheng, W., Huang, T., Cheng, K., Gudovskiy, D., Okuno, T., Nakata, Y., Keutzer, K., et al. Sparsevlm: Visual token sparsification for efficient vision-language model inference. *arXiv preprint arXiv:2410.04417*, 2024c.
- Zhao, H., Liu, Z., Liu, Y., Qin, Z., Liu, J., and Gedeon, T. Facephi: Lightweight multimodal large language model for facial landmark emotion recognition. In *5th Workshop on practical ML for limited/low resource settings*, 2024.
- Zhao, J., Wei, X., and Bo, L. R1-omni: Explainable omni-multimodal emotion recognition with reinforcement learning. *arXiv preprint arXiv:2503.05379*, 2025.
- Zhu, J., Wang, W., Chen, Z., Liu, Z., Ye, S., Gu, L., Tian, H., Duan, Y., Su, W., Shao, J., Gao, Z., Cui, E., Wang, X., Cao, Y., Liu, Y., Wei, X., Zhang, H., Wang, H., Xu, W., Li, H., Wang, J., Deng, N., Li, S., He, Y., Jiang, T., Luo, J., Wang, Y., He, C., Shi, B., Zhang, X., Shao, W., He, J., Xiong, Y., Qu, W., Sun, P., Jiao, P., Lv, H., Wu, L., Zhang, K., Deng, H., Ge, J., Chen, K., Wang, L., Dou, M., Lu, L., Zhu, X., Lu, T., Lin, D., Qiao, Y., Dai, J., and Wang, W. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models, 2025. URL <https://arxiv.org/abs/2504.10479>.